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ORIGINAL ARTICLE

Numerical simulation of the Vasilev reflection

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Abstract We present a high resolution numerical solution of the Vasilev reflection configuration within the framework of depth averaged two-dimensional inviscid shallow water flow. The study provides the details of the steady flow field and shock wave pattern close to the triple point which confirm the four-wave theory. The shape of the reflected shock in the region upstream of the supercritical patch is also investigated.

Keywords Vasilev reflection · Weak shock reflection · Four-wavetheory · Supercritical channel flow

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1 Introduction

In steady supercritical channel flow the reflection of a shock wave over a straight wall leads to three main, distinctly different wave configurations namely, regular and irregular reflection, whose solution is provided by the two- and the three-shock theory developed by von Neumann [11], and Guderley and Vasilev reflections, whose solution is provided by the four-wave theory suggested by Guderley [2].

In gas dynamics, the Guderley reflection (GR) has been simulated numerically at moderately low Mach numbers. The numerical simulations performed by Vasil'ev [8], Vasil'ev and Kraiko [9], and Zakharian et al. [12] using the full Euler equations and the numerical solution of the unsteady transonic disturbance equations for an ideal gas provided by Hunter and Brio [3] and Tesdall and Hunter [5], all confirmed the validity of the four-wave theory.

Recently, the shock wave pattern predicted by the four-wave theory has been further confirmed by Skews and Ashworth [4] who experimentally detected the expansion fan and the supersonic patch behind the triple point using a special shock tube.

In the Vasilev reflection (VR) the solution is again provided by the four-wave theory but, unlike the GR, the flow between the slip stream and the Mach stem is subsonic (or subcritical, within the framework of shallow water flow) [1].

The main difficulty in reproducing a VR, either numerically or experimentally, stems from the relatively high Mach number (or Froude number) which characterizes this reflection configuration. In fact, at moderately high Mach number the supercritical patch behind the triple point is awfully small, so that it requires an extremely refined computational grid to be resolved.

In this work, we resolve a VR numerically to provide new information about this reflection configuration. We give the details of the numerical approach and we discuss in depth the results.

The solution is obtained within the framework of depth averaged two-dimensional inviscid shallow water flow. However, because of the strong analogy between shallow water flow and either two-dimensional gas flow or shallow granular flow, we are confident that the present work can be of more general interest.

2 The Vasilev reflection

The shock wave pattern of the VR is very similar to that of the irregular (or Mach) reflection (see Fig. 1). The incident

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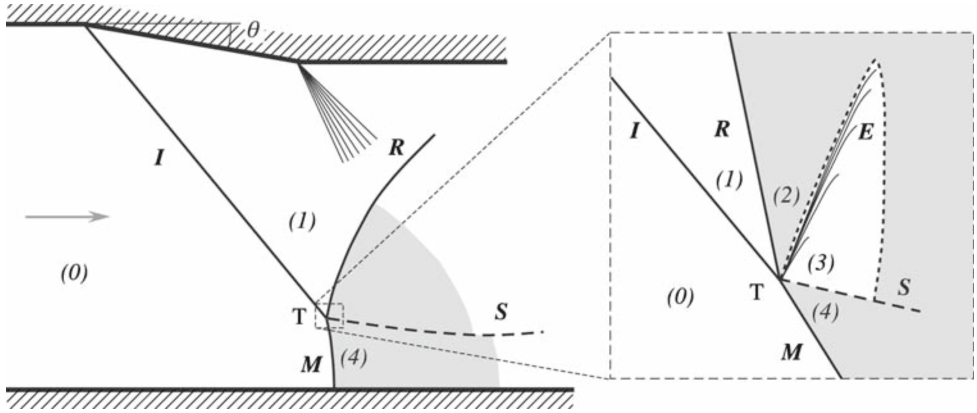


Fig. 1 Steady flow in a linear channel contraction producing a Vasilev reflection. The *shaded area* has $F < 1$, the flow is from left to right

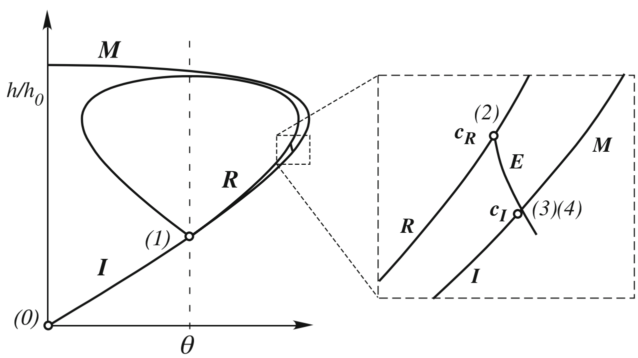


Fig. 2 Shock polar diagram of the overall and local triple point structure for the Vasilev reflection shown in Fig. 1. c_R and c_I denote the critical condition for the reflected and incident shock polar, respectively

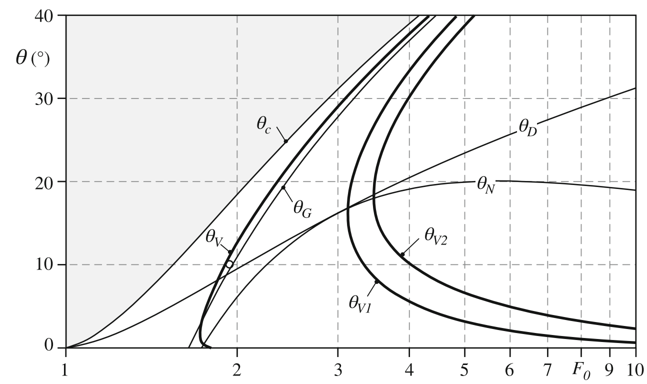


Fig. 3 Reflection domain in the (F_0, θ)-plane. Vasilev reflection is possible for flow conditions between the curves θ_V and θ_{V1}. In the domain region between curves θ_{V1} and θ_{V2} two distinct Vasilev reflection configurations are possible. For the sake of clarity, the detachment condition θ_D and the normal condition θ_N are also plotted. No reflection is possible for $\theta > \theta_c$. For $\theta_G < \theta < \theta_c$ and $\theta > \theta_D$ the four-wave theory only provides physical solutions. The *circle* denotes the reflection condition which is simulated with the numerical model: $F_0 = 1.93$, $\theta = 10°$

shock I, the reflected shock R, and the Mach stem M, meet at the triple point T. A contact discontinuity, the slip stream S, and an expansion fan (E) also originate at T. Flow velocity is discontinuous across S, while water depth and flow direction are both continuous. The expansion fan accelerates the flow from critical condition (sonic condition in gas dynamics) behind the reflected shock and produces a sail-shaped patch of supercritical flow (supersonic flow in gas dynamics) attached to the slip stream and embedded inside the subcritical flow (subsonic flow in gas dynamics) downstream of the reflected shock and the Mach stem (Fig. 1).

The Vasilev reflection can be regarded as a special GR in that the solution is provided by the four-wave theory [1,10]. It differs from the GR because the flow between the slip stream and the Mach stem is subcritical. In the shock polar representation, the VR occurs when the characteristic from the critical point c_R of the reflected shock polar intersects the subcritical branch of the incident shock polar (Fig. 2).

2.1 The Vasilev reflection domain

Figure 3 shows the VR domain in the (F_0, θ)-plane, F_0 being the Froude number of the incoming flow. The limit condition which separates the VR from the GR is the deviation θ_V such that the characteristic E from point c_R (Fig. 2) intersects the incident shock polar exactly at the critical condition c_I.

A single Vasilev solution exists when flow conditions lay in the region of the reflection domain bounded by the curves θ_V and θ_{V1}, θ_{V1} being the flow deviation such that the incident and the reflected shock polars intersects at the critical condition c_R (upper panel of Fig. 4). Two distinct VRs are possible when flow conditions lay between curves θ_{V1} and θ_{V2}. The latter is defined as the flow deviation such that the characteristic E from point c_R is externally tangent to the incident shock polar (lower panel of Fig. 4).

Vasilev reflection is the only solution when flow conditions lay in the region of the reflection domain bounded by the curves θ_V, θ_D, and θ_G. The limit condition θ_G occurs when the reflected shock polar is internally tangent to the incident shock polar.

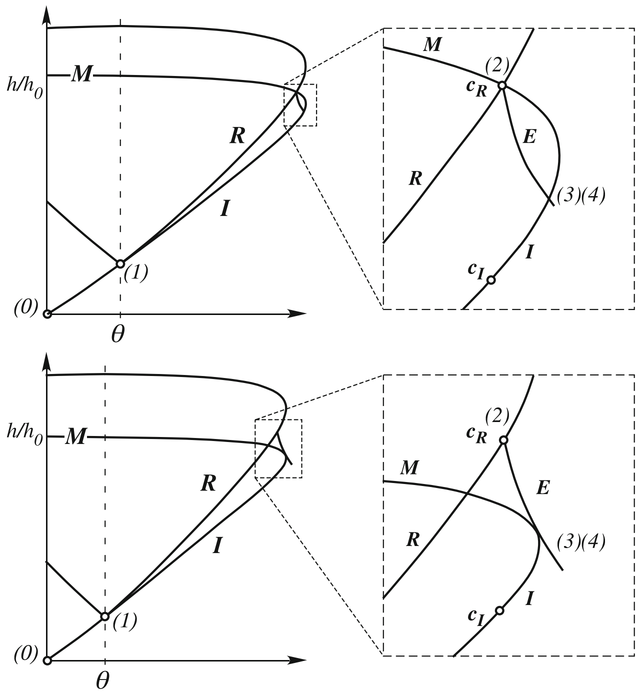


Fig. 4 Shock polar representation of limit conditions for the Vasilev reflection: θ_{V1} condition (*upper panel*) and θ_{V2} condition (*lower panel*)

We recall here that the above boundaries of the VR domain have been determined within the framework of shallow water flow, and they only qualitatively compare with those discussed in [1,10] for the case of gas flow.

2.2 The numerical model

The unsteady depth-averaged shallow water equations (SWE) are used to solve for the VR pattern in a supercritical channel flow with a linear contraction at moderately low Froude number.

The conservative form of depth-averaged SWE is written in vector form as

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \mathbf{F} = \mathbf{S} \tag{1}$$

where

$$\mathbf{U} = \begin{pmatrix} h \\ uh \\ vh \end{pmatrix}, \quad \mathbf{F} = \begin{pmatrix} hu & hv \\ hu^2 + \frac{1}{2}gh^2 & huv \\ huv & hv^2 + \frac{1}{2}gh^2 \end{pmatrix} \tag{2}$$

and u, v are the flow velocity components in x, y directions, g is gravity, and h is the flow depth. On neglecting viscosity and assuming an horizontal bottom, the source term $\mathbf{S}$ in Eq. 1 is identically zero.

The SWE are solved with a second-order accurate, Godunov-type, finite volume method on an unstructured triangular grid. We apply weighted averaged flux (WAF) scheme [6] and remove spurious oscillations in the vicinity of high water depth and velocity gradients by enforcing a total variation diminishing (TVD) constraint to the scheme. The chosen limiter function is equivalent to van Leer's conventional flux limiter [7].

In order to resolve the solution near the triple point we use a nested-block grid refinement technique which consists of two steps.

In the first step the steady-state solution is computed over the whole domain with appropriate boundary conditions. In this step we use a very refined grid along the shocks and over the subcritical flow region behind the Mach stem and the reflected shock.

In the second step only a portion of the domain around the triple point is considered. This sub-domain is meshed with a finer resolution grid and the steady-state solution is then computed with initial and boundary conditions extracted from the converged results of the preceding simulation.

The second step is iterated as many times as necessary until the required resolution is achieved. In each iteration the number of computational elements is maintained and the domain is reduced approximately by a factor of four. As a result, the grid size is halved at each step.

It is worth noting that standard local grid refinement techniques implicitly assume that inaccuracy of the solution over the coarsely discretized part of the domain weakly affects the solution over the refined region. The present numerical technique further assumes that the symmetric statement holds, i.e., that the increased accuracy of the solution over the refined grid has a negligible impact on the solution over the neighboring coarser grid. A grid refinement study confirmed that this assumption is fairly reasonable.

We simulate the shock reflection pattern in a steady supercritical channel flow with a linear contraction (Fig. 1). The incoming flow has $F_0 = 1.93$ and it is deviated by an angle $\theta = 10°$. This flow condition lays within the VR domain and is marked by a circle in Fig. 3.

The starting grid size Δ around the triple point, along the shocks and in the subcritical flow region behind the Mach stem and the reflected shock is approximately 0.03 h_M, h_M being the Mach stem height.

The downstream boundary is situated far enough from the channel contraction so that the flow is supercritical on it and a radiation boundary condition is prescribed. At the upstream boundary, where the flow is uniformly supercritical, all variables (h, u, $v = 0$) are specified. The upper and lower boundaries (i.e., the channel walls) are assumed impermeable and a free slip condition is imposed.

Figure 5 shows the leading order solution computed over the whole domain. The flow behind the reflected shock and the Mach stem close to the triple point is fully subcritical and the existence of a supercritical patch cannot be envisaged.

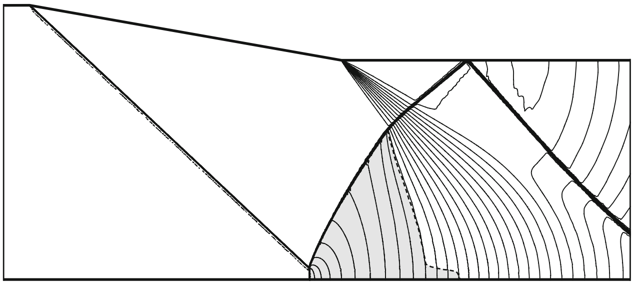


Fig. 5 The leading order computational domain and the numerical solution for $F_0 = 1.93$ and $\theta = 10°$: *solid lines* are h/h_0-contours (contour spacing is 0.04), *dashed line* is the critical line. The *shaded area* has $F \leq 1$. The flow is from left to right

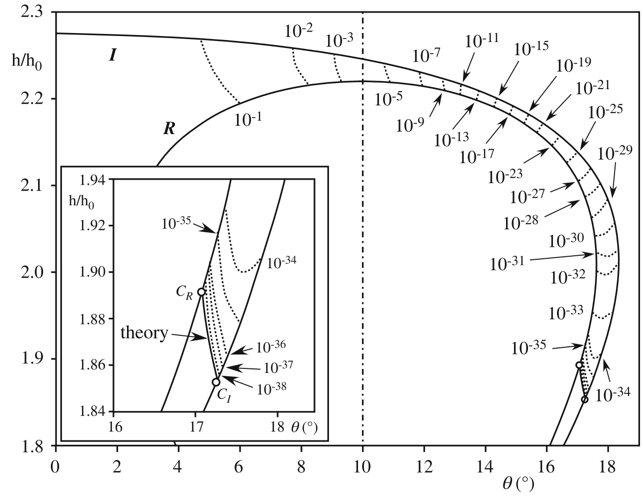


Fig. 6 Shock polar representation of the Vasilev reflection for $F_0 = 1.93$ and $\theta = 10°$. *Dotted lines* show the computed flow depth and flow direction extracted, at each grid resolution, along a circle of radius $r_0 \approx 10\Delta$ with center point on the triple point. *Labels* denote the non dimensional radius r_0/h_M

2.3 Results and discussion

The computation requires a large number of refining steps to resolve the four-wave pattern close to the triple point because of the relatively high Froude number. The most refined grid has a mesh size $\Delta = 1.55 \cdot 10^{-39} h_M$.

We track the solution convergence, at each computational iteration, by plotting on the polar diagram the computed flow depth and flow direction extracted along a circle of radius $r_0 \approx 10\Delta$ with center point on the triple point. The solution in the subcritical region behind the reflected shock and the Mach stem is given by the dotted curves connecting the incident shock polar to the reflected shock polar shown in Fig. 6. The curves actually plotted in Fig. 6 have been interpolated from the original solution to improve legibility. It can be observed that the convergence velocity reduces where the incident and reflected shock polars are close to each other and in the vicinity of the converged solution (i.e., four wave solution).

Figure 7 shows the shock pattern close to the triple point at different resolutions. As the grid size and thus the distance from the triple point reduce the flow field experiences striking changes: the iso-Froude contours in frames *a* to *d* have completely different patterns. The reflected shock and the Mach stem both rotate counterclockwise while the slip stream experiences an opposite rotation. The Froude number behind the reflected shock and in the flow region between the slip stream and the Mach stem slowly increases toward critical condition (frames *a* to *e*).

A small, poorly resolved supercritical patch is detected behind the triple point when the grid size is approximately $\Delta = 1.1 \cdot 10^{-35} h_M$ (Fig. 7d). At this zoom level, the iso-Froude contours behind the triple point take a pattern which resemble the shape of the supercritical patch.

After the appearance of the supercritical patch iso-Froude contours maintain similar patterns as the refinement procedure progresses.

The patch of supercritical flow behind the triple point takes a well defined shape at a grid size $\Delta = 6.8 \cdot 10^{-37} h_M$ and the height of the patch is approximately $5 \cdot 10^{-35} h_M$ (Fig. 7e).

Details of the flow and shock pattern close to the triple point are given in Fig. 7f. The reflected shock, the Mach stem, the slip stream and the expansion fan have the directions predicted by the four-wave theory. The flow is subcritical below the slip stream with a Froude number ranging from barely smaller than one close to the triple point to $F \approx 0.97$ at the downstream end of the flow region below the supercritical patch. The computed Froude number in the supercritical patch is $F \approx 1.02$, slightly smaller than the theoretical value $F = 1.03$.

Figure 8 shows the flow depth and flow direction at the same resolution of frames *e* and *f* of Fig. 7.

In Fig. 8a, the relative flow depth across the supercritical patch has a behavior very similar to that of Froude number at the same resolution. Flow depth smoothly increases downstream away from the expansion fan and it reaches the value $h/h_0 \approx 1.896$ along the critical line bounding downstream the supercritical patch. However, as opposite to iso-Froude contours, iso-levels do not kink at the lower boundary of the supercritical patch because flow depth is continuous across the slip stream.

Close to the triple point (Fig. 8c), the iso-h/h_0 contours show a complex pattern. However, because of the very small water depth differences (iso-h/h_0 contours have a line spacing of 0.004), this result might be a numerical artifact due to spurious oscillations.

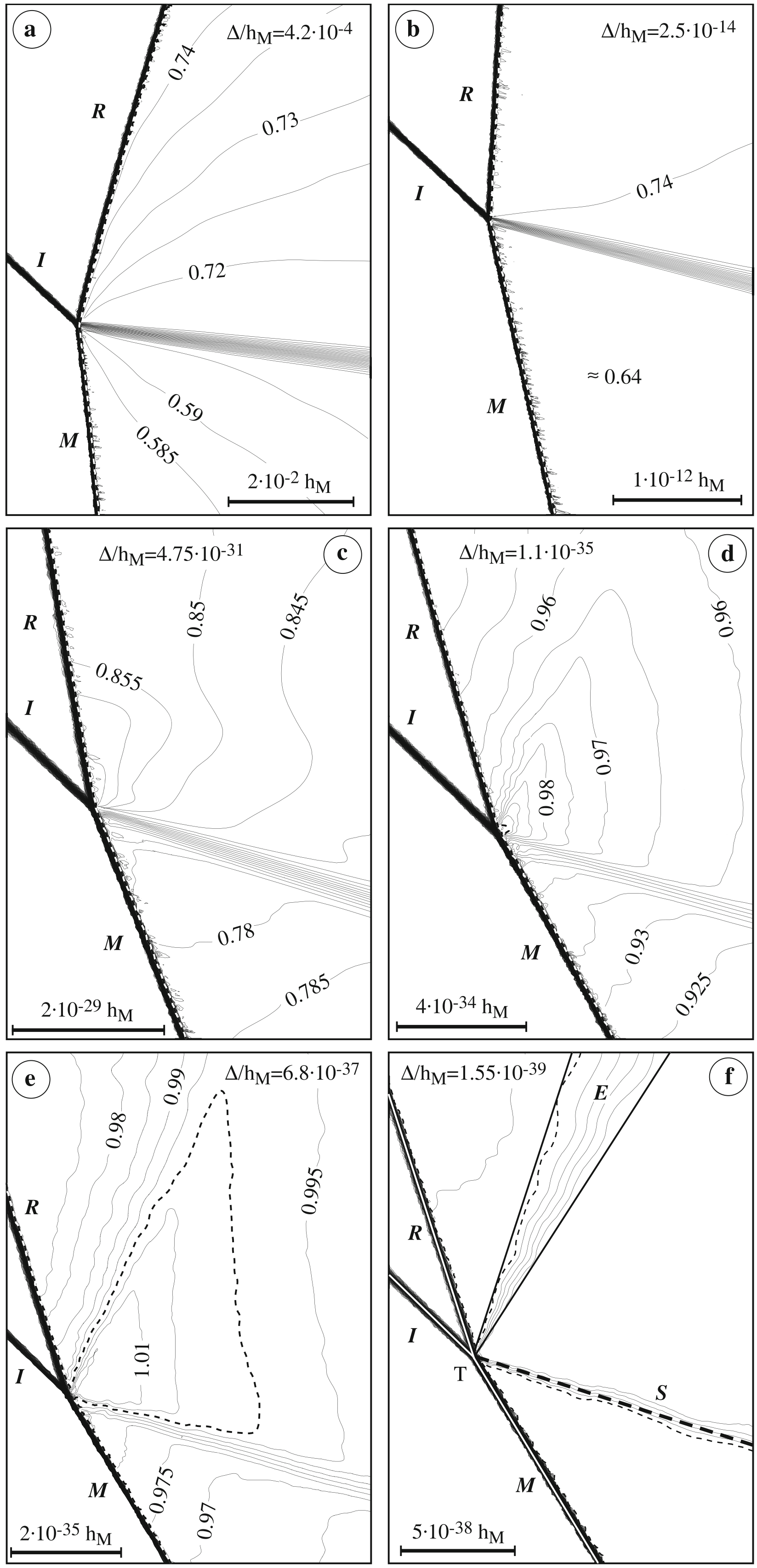


Fig. 7 The Vasilev reflection pattern close to triple point T for $F_0 = 1.93$ and $\theta = 10°$ at different grid resolutions (i.e., at different zoom levels). The *solid lines* are iso-Froude contours (line spacing is 0.005); the *thin dashed line* is the critical line. In the last frame (*f*) the analytical solution of the four-wave model is superimposed to the numerical solution: the *white lines* are the Mach stem, the incident and the reflected shock, the *thin solid lines* enfold the expansion fan E, and the thick dashed line is the slipstream

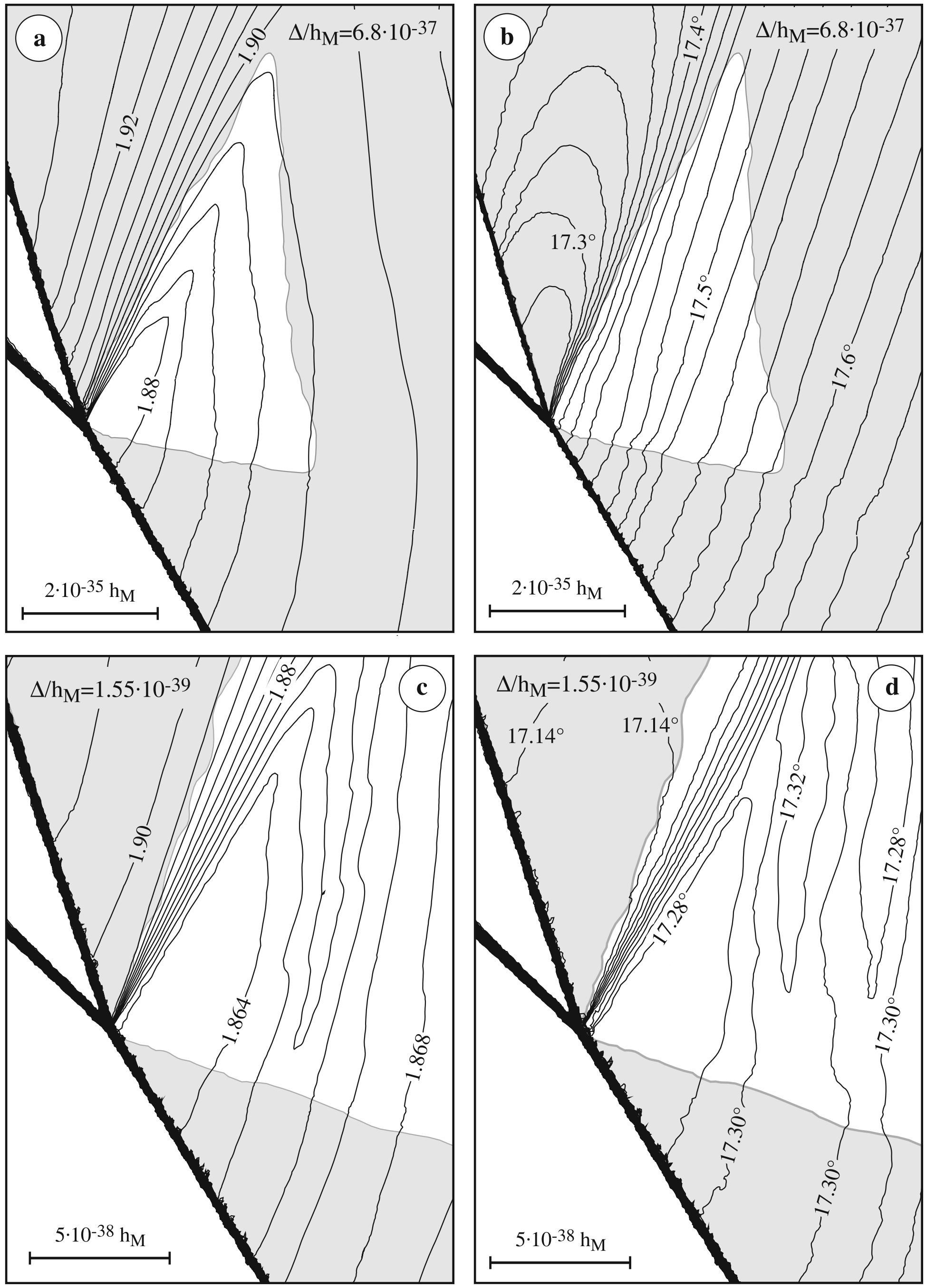


Fig. 8 The Vasilev reflection pattern close to the triple point for $F_0 = 1.93$ and $\theta = 10°$ at different grid resolutions (i.e., at different zoom levels). The *solid lines* are iso-h/h_0 contours in (a) and (c) (line spacing is 0.004), and iso-direction contours in (b) and (d) (line spacing is 0.02°, angles are given with respect to the horizontal and are positive when the flow is toward the reflecting wall). The *shaded area* has $F \leq 1$

The flow direction has a quite different behavior (Fig. 8b). The iso-θ contours are approximately straight lines across the supercritical patch and have approximately the same direction as the expansion fan. Again, because of the very small direction differences (iso-θ contours have a line spacing of 0.02°), the complex pattern of the flow direction inside the region downstream of the reflected shock and the Mach stem, and very close to the triple point (Fig. 8d) is possibly due to numerical oscillations.

Numerical results suggest that a weak and diffused compression wave affects the flow in the supercritical patch. In fact, water depth in the supercritical patch increases gradually from the end of the expansion fan towards the downstream critical line (Fig. 8a). In addition, the flow in the supercritical patch gently turns toward the reflecting wall (Fig. 8b) suggesting that the compression wave originates from along the upper boundary of the patch, i.e. from the critical line bounding downstream the patch. Here, a vanishingly small deviation toward the reflecting wall is imposed by the subcritical flow immediately downstream, which possibly triggers the compression wave.

Quantitative comparison between four-wave theory and numerical results close to the triple point is given in Table 1. Apart from the moderate difference between the theoretical and numerical relative water depth and flow direction inside the supercritical patch (the error is smaller than 0.5%), the overall agreement between theory and computational results is good and confirms the validity of the four-wave theory.

The present numerical technique allows for a detailed reconstruction of the shape of the reflected shock. At each refinement step, the simulated branch of the reflected shock is extracted at discrete points (approximately 15 ∼ 20 points).

The computed angle β between the reflected shock and the x-axis is plotted in Fig. 9, where the chosen coordinate system has the origin at the triple point and is oriented such that the y-axis is tangent to the reflected shock at the origin (inset of Fig. 9).

The scatter of the plotted points is due to the finite width of the numerical reflected shock front (approximately $2\Delta/3\Delta$). This introduces some inaccuracy in the tracking of the front position and in the evaluation of the triple point position.

Table 1 Comparison between four-wave theory and numerical results close to the triple point

	Upstream of the reflected shock	Inside the supercritical patch	Below the slipstream
h/h_0	1.893 (*1.896*)	1.855 (*1.864*)	1.855 (*1.864*)
F	1.00 (∼*1.00*)	1.03 (*1.02*)	0.998 ($\lesssim$*1.00*)
θ	17.07° (*17.14°*)	17.26° (*17.28°*)	17.26° (*17.28°*)

Italic is used to denote numerical solution

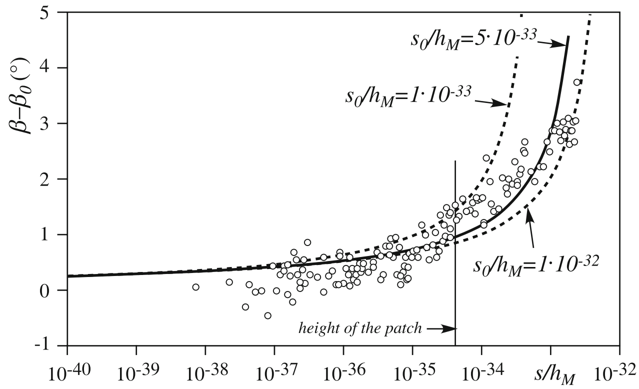


Fig. 10 Comparison between theoretical (Eq. 3) and numerical shape of the reflected shock front close to the triple point

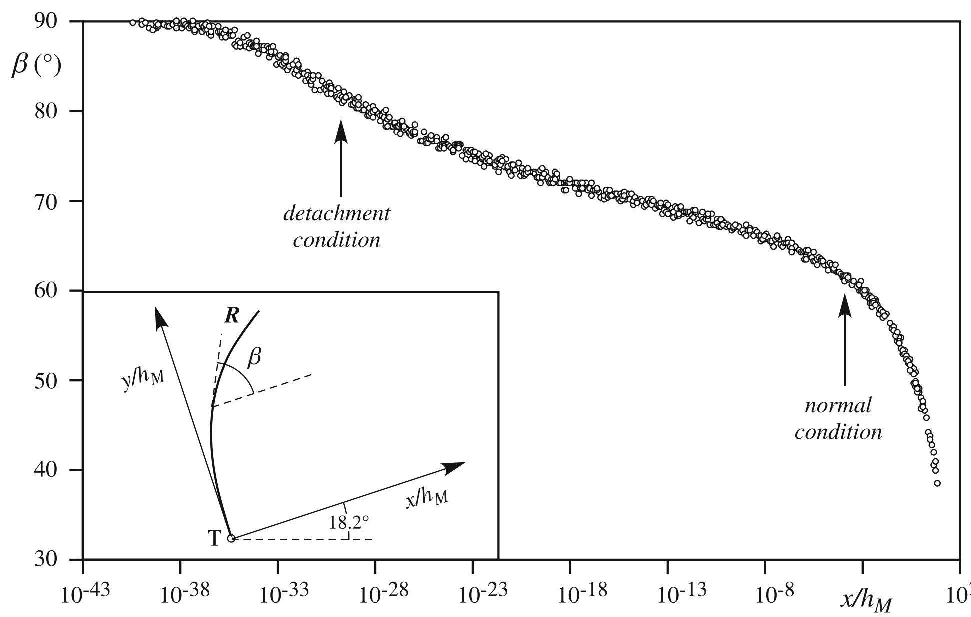
Vasil'ev [8] developed a simple theory to analytically predict the shape of the reflected shock front in the vicinity of the triple point. Assuming inviscid flow conditions, neglecting the energy dissipation across the reflected shock, and with a few additional minor assumptions, he proposed the following law which is here rewritten within the framework of shallow water flow

$$\beta = \beta_0 - \frac{\sqrt{F_1 - 1}}{8\ln(s/s_0)} \tag{3}$$

Here F_1 is the Froude number upstream of the reflected shock, s is the curvilinear coordinate along the reflected front with origin at the triple point, s_0 is a length scale, β is the reflected shock angle and $\beta_0 = \beta(s \to 0)$. According to [8], Eq. 3 applies for $s \ll s_0$.

Figure 10 compares present numerical results and Eq. 3 with s_0 in the range $(10^{-33} - 10^{-32})h_M$. The agreement between the theory and the numerical results is not fully

Fig. 9 The angle β of the reflected shock as a function of the distance x/h_M from the triple point. The *arrows* indicate the position of the normal shock and the detachment condition for the reflected shock



satisfactory. However, our numerical results do not possess the sufficient accuracy to definitely evaluate the simplified theory proposed by Vasilev.

2.4 Conclusions

We have presented numerical evidence of the structure of the reflected shock, expansion fan and supercritical patch in a tiny region behind the triple point of an inviscid VR.

Numerical results show that the four-wave theory correctly predicts the flow and shock wave pattern close to the triple point of this reflection configuration.

We have shown that the flow in the supercritical patch is not uniform, but it is affected by a weak and diffused compression wave. Numerical results suggest that this compression wave originates from the interaction of the supercritical flow in the patch with the subcritical flow immediately downstream, which imposes a vanishingly small deviation toward the reflecting wall at the critical line.

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